

6.3 Solution: (a) Let

$$\mathbf{A}(\mathbf{x}, t) = \int \frac{d^3 k}{(2\pi)^3} \mathbf{A}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}},$$

we will have

$$\nabla^2 \mathbf{A}(\mathbf{x}, t) = - \int \frac{d^3 k}{(2\pi)^3} k^2 \mathbf{A}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}}.$$

Then, Eq. (5.160) becomes

$$\mu\sigma \int \frac{d^3 k}{(2\pi)^3} \frac{\partial \mathbf{A}(\mathbf{k}, t)}{\partial t} e^{i\mathbf{k} \cdot \mathbf{x}} = - \int \frac{d^3 k}{(2\pi)^3} k^2 \mathbf{A}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}},$$

which must be valid for each $\mathbf{k}$, *i.e.*,

$$\frac{\partial \mathbf{A}(\mathbf{k}, t)}{\partial t} = - \frac{k^2}{\mu\sigma} \mathbf{A}(\mathbf{k}, t).$$

With the initial condition

$$\mathbf{A}(\mathbf{k}, 0) = \int \mathbf{A}(\mathbf{x}, 0) e^{-i\mathbf{k} \cdot \mathbf{x}} d^3 x,$$

the solution to the first order ODE for $\mathbf{A}(\mathbf{k}, t)$ is

$$\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k}, 0) e^{-k^2 t / \mu\sigma},$$

for $t > 0$. Therefore,

$$\begin{aligned} \mathbf{A}(\mathbf{x}, t) &= \int \frac{d^3 k}{(2\pi)^3} \mathbf{A}(\mathbf{k}, 0) e^{-k^2 t / \mu\sigma} e^{i\mathbf{k} \cdot \mathbf{x}} \\ &= \int \frac{d^3 k}{(2\pi)^3} \left(\int d^3 x' \mathbf{A}(\mathbf{x}', 0) e^{-i\mathbf{k} \cdot \mathbf{x}'} \right) e^{-k^2 t / \mu\sigma} e^{i\mathbf{k} \cdot \mathbf{x}} \\ &= \int d^3 x' \left(\int \frac{d^3 k}{(2\pi)^3} e^{-k^2 t / \mu\sigma} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \right) \mathbf{A}(\mathbf{x}', 0) \\ &= \int d^3 x' G(\mathbf{x} - \mathbf{x}', t) \mathbf{A}(\mathbf{x}', 0), \end{aligned}$$

with the Green function

$$G(\mathbf{x} - \mathbf{x}', t) = \int \frac{d^3 k}{(2\pi)^3} e^{-k^2 t / \mu\sigma} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')}.$$

(b) Perform the Fourier transform on the Green function obtained in part (a) in space and time, we have

$$\begin{aligned} G(\mathbf{k}, \omega) &= \int d^3 x \int_0^{+\infty} dt G(\mathbf{x} - \mathbf{x}', t) e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}') + i\omega t} \\ &= \int d^3 x \int_0^{+\infty} dt \int \frac{d^3 k_1}{(2\pi)^3} e^{-k_1^2 t / \mu\sigma} e^{i\mathbf{k}_1 \cdot (\mathbf{x} - \mathbf{x}')} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}') + i\omega t} \\ &= \int_0^{+\infty} e^{i\omega t - k^2 t / \mu\sigma} dt \\ &= \frac{1}{-i\omega + \frac{k^2}{\mu\sigma}}. \end{aligned}$$

Then,

$$G(\mathbf{x} - \mathbf{x}', t) = \int \frac{d^3k}{(2\pi)^3} \int \frac{d\omega}{2\pi} G(\mathbf{k}, \omega) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}') - i\omega t} = \int \frac{d^3k}{(2\pi)^3} \int \frac{d\omega}{2\pi} \frac{e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}') - i\omega t}}{-i\omega + \frac{k^2}{\mu\sigma}},$$

and

$$\left(\frac{\partial}{\partial t} - \frac{1}{\mu\sigma} \nabla^2 \right) G = \int \frac{d^3k}{(2\pi)^3} \int \frac{d\omega}{2\pi} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}') - i\omega t} = \delta^{(3)}(\mathbf{x} - \mathbf{x}') \delta(t).$$

(c) From part (a),

$$\begin{aligned} G(\mathbf{x} - \mathbf{x}', t) &= \frac{1}{4\pi^2} \int_0^{+\infty} k^2 dk \int_{-1}^1 d(\cos \theta) e^{ik|\mathbf{x} - \mathbf{x}'| \cos \theta - k^2 t / \mu\sigma} \\ &= \frac{1}{4\pi^2} \int_0^{+\infty} \frac{e^{ik|\mathbf{x} - \mathbf{x}'|} - e^{-ik|\mathbf{x} - \mathbf{x}'|}}{ik|\mathbf{x} - \mathbf{x}'|} e^{-k^2 t / \mu\sigma} k^2 dk \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \frac{e^{ik|\mathbf{x} - \mathbf{x}'| - k^2 t / \mu\sigma}}{i|\mathbf{x} - \mathbf{x}'|} k dk \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \frac{k}{i|\mathbf{x} - \mathbf{x}'|} \exp \left(-\frac{t}{\mu\sigma} \left[k - \frac{i\mu\sigma}{2t} |\mathbf{x} - \mathbf{x}'| \right]^2 - \frac{\mu\sigma}{4t} |\mathbf{x} - \mathbf{x}'|^2 \right) dk \\ &= \frac{1}{4\pi^2} \exp \left(-\frac{\mu\sigma}{4t} |\mathbf{x} - \mathbf{x}'|^2 \right) \int_{-\infty}^{+\infty} \left(\frac{k}{i|\mathbf{x} - \mathbf{x}'|} + \frac{\mu\sigma}{2t} \right) e^{-k^2 t / \mu\sigma} dk \\ &= \frac{1}{4\pi^2} \exp \left(-\frac{\mu\sigma}{4t} |\mathbf{x} - \mathbf{x}'|^2 \right) \frac{\mu\sigma}{2t} \sqrt{\frac{\mu\sigma\pi}{t}} \\ &= \left(\frac{\mu\sigma}{4\pi t} \right)^{3/2} \exp \left(-\frac{\mu\sigma}{4t} |\mathbf{x} - \mathbf{x}'|^2 \right), \end{aligned}$$

for $t > 0$. Therefore, the final form of the Green function is

$$G(\mathbf{x}, t; \mathbf{x}', 0) = \Theta(t) \left(\frac{\mu\sigma}{4\pi t} \right)^{3/2} \exp \left(-\frac{\mu\sigma}{4t} |\mathbf{x} - \mathbf{x}'|^2 \right).$$

This is nothing but just a heat kernel in three-dimensional Euclidean geometry.